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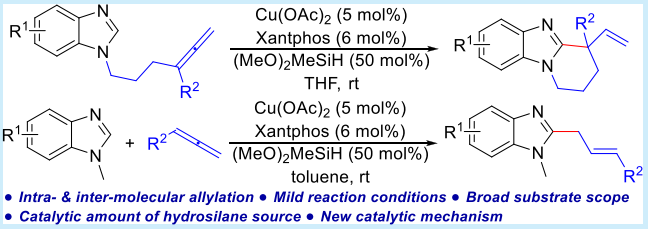
Cu-Catalyzed C–H Allylation of Benzimidazoles with Allenes

Yaxi Dong and Bernhard Breit*

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ABSTRACT: CuH-catalyzed intramolecular cyclization and intermolecular allylation of benzimidazoles with allenes have been described. The reaction proceeded smoothly with the catalytic system of $Cu(OAc)_2$/Xantphos and catalytic amount of $(MeO)_2MeSiH$. This protocol features mild reaction conditions and a good tolerance of substrates bearing electron-withdrawing, electron-donating, or electron-neutral groups. A new catalytic mechanism was proposed for this copper hydride catalytic system.

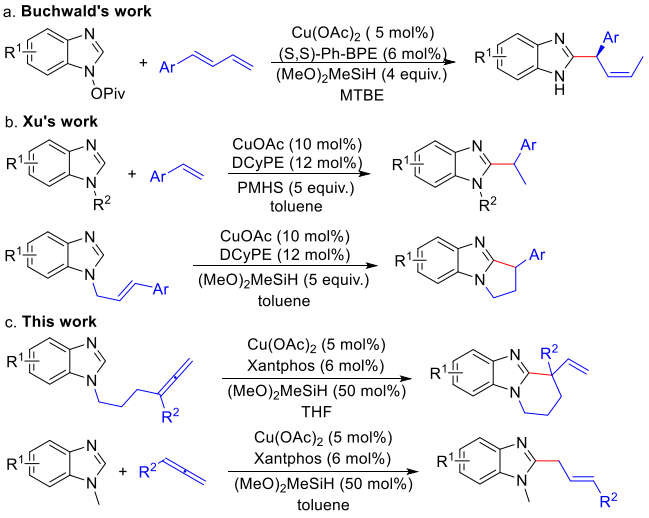


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Functionalization of benzimidazoles is an important topic because these scaffolds widely occur in pharmaceuticals, natural products, and functional materials.[1] For quite a long time, the functionalization of benzimidazoles has been enriched by classical metalation with strong bases.[2] However, these traditional synthetic methods suffer general limitations, such as harsh reaction conditions, low chemo- and/or regioselectivity, and poor reactivity of electron-deficient benzimidazoles. In recent decades, direct transition-metal-catalyzed C–H functionalization has been an effective tool for the synthesis of benzimidazole frameworks. For examples, in 2001, Bergman and Ellman reported a Rh-catalyzed intramolecular cyclization of benzimidazles.[3] Cavell, Ye, and Ackermann independently described the intramolecular cyclization of benzimidazoles using nickel catalysts.[4] On the other hand, there are some reported examples of intermolecular C–H functionalization of benzimidazoles that rely on alkylation, alkenylation, or alkynylation of benzimidazoles using rhodium, palladium, or nickel catalysts.[5−7] However, these methods still have many challenges, such as the need of a strong base, high temperature, and limited scope of benzimidazoles. Therefore, the development of effective synthetic methodologies is undoubtedly urgent and challenging.

In the past few years, CuH catalysis has been widely applied to the hydrofunctionalization of alkenes, alkynes, and ketones with various electrophiles.[8] However, only a few electron-deficient heterocycles have been studied. For examples, Buchwald and Ge independently reported several examples of CuH-catalyzed C–H alkylation of N-heterocycles, e.g., pyridines, indazoles, and quinolines.[9] Recently, Buchwald and co-workers described a CuH-catalyzed branch-selective alkylation of benzimidazoles with 1,3-dienes (Scheme 1a).[10] Xu and co-workers developed a CuH-catalyzed branch-selective alkylation of benzimidazoles with alkenes (Scheme 1b).[11] The typical CuH catalytic process undergoes a hydrocupration of alkene followed by the addition to the C=N bond and regeneration of CuH species with an over-stoichiometric amount of hydrosilane. Our group has described a Rh- and Pd-catalyzed regiodivergent and stereoselective N allylation of imidazoles with allenes.[12] In line with our continued interest in functionalization of benzimidazoles, herein, we report the first example of CuH-catalyzed C2 allylation of benzimidazoles with allenes (Scheme 1c).

Scheme 1. Functionalization of Benzimidazoles



The initial experiments were performed with N1-tethered allene benzimidazole **1a**–**1c** with the typical CuH catalytic conditions (entries 1–5 of Table 1). To our delight, Xantphos

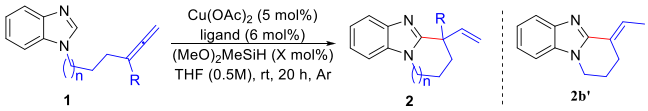
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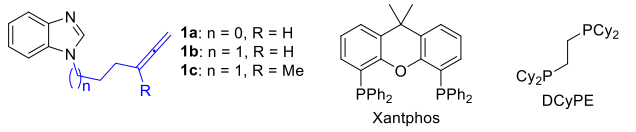
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Table 1. Conditions Optimization for CuH-Catalyzed Cyclization of Benzimidazles[a]



entry	1	X	ligand	yield (%)[b]
1	**1a**	200	DCyPE	0
2	**1a**	200	Xantphos	0
3	**1b**	200	DCyPE	0
4[c]	**1b**	200	Xantphos	54
5	**1c**	200	Xantphos	>99
6	**1c**	100	Xantphos	>99
7	**1c**	50	Xantphos	>99
8	**1c**	40	Xantphos	74
9	**1c**	30	Xantphos	trace
10[d]	**1c**	50	Xantphos	61

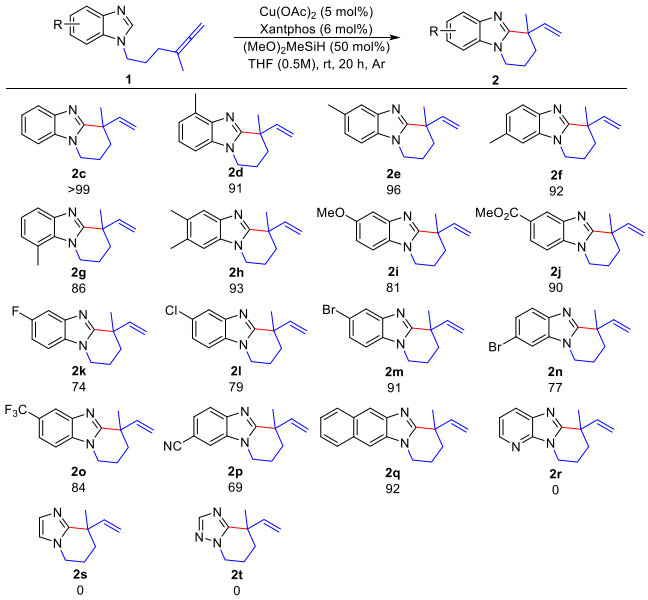
[a]Reaction conditions: compound **1** (0.4 mmol), $Cu(OAc)_2$ (5 mol %), ligand (6 mol %), and THF (2 mL), at room temperature (rt) under argon for 20 h. [b]Isolated yield. [c]Compound **2b′** yield = 16%. [d]Using $(EtO)_2MeSiH$ as the hydrosilane source.



was found to give the desired product **2b** in a 54% yield, while a byproduct **2b′** was obtained in a 16% yield (entry 4 of Table 1). Especially, the desired product **2c** was obtained in an excellent yield (entry 5 of Table 1). Subsequently, the reaction of benzimidazole **1c** was selected as the model reaction, and the effect of hydrosilane was investigated. Interestingly, the use of 50 mol % of $(MeO)_2MeSiH$ also gave the desired product **2c** in an excellent yield (entry 7 of Table 1), which suggested that this reaction perhaps followed a different catalytic mechanism compared to a previous classic copper hydride catalytic mechanism.[10,11] However, a further decrease of the amount of $(MeO)_2MeSiH$ decreased the yield significantly (entries 8 and 9 of Table 1). Another commonly used hydrosilane source, $(EtO)_2MeSiH$, was also checked. Unfortunately, it only provided the desired product in a 61% yield (entry 10 of Table 1). Further examination of various catalysts, solvents, and ligands exhibited lower activity (see the Supporting Information for details).

With the optimized reaction conditions in hand, we next investigated the applicability of this cyclization protocol for various benzimidazoles (Scheme 2). The results demonstrated that benzimidazoles bearing electron-donating, electron-withdrawing, or electron-neutral groups were all well-tolerated in this reaction, providing the corresponding products in moderate to excellent yields (**2c**−**2q**). Benzimidazoles bearing a methyl substituent at different positions afforded the corresponding products in 86−96% yields (**2d**−**2g**). Good yields were also obtained for other benzimidazoles containing electron-donating groups, e.g., 5,6-dimethyl and 5-methoxy (**2h** and **2i**). Ester-containing substrate could provide the desired product in a 90% yield (**2j**). Substrates containing halogen groups (e.g., F, Cl, and Br) could also survive in this CuH-catalyzed cyclization reaction, affording the corresponding products in moderate to good yields (**2k**−**2n**). Other substrates bearing electron-withdrawing groups (e.g., CF_3 and CN) were also well-tolerated in this reaction (**2o** and **2p**). A

Scheme 2. Substrate Scope of CuH-Catalyzed Allylation of Benzimidazoles[a,b]

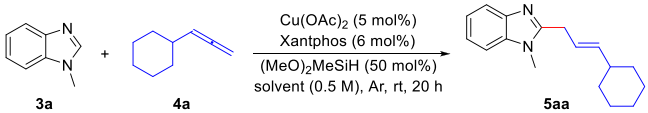


[a]Reactions were carried out with reaction conditions in entry 7 of Table 1. [b]Isolated yield.

good yield was also obtained for electron-neutral benzimidazoles (**2q**). However, N-containing heterocycles, imidazoles, or triazoles tethered to allene could not provide the corresponding products (**2r**−**2t**).

Encouraged by this cyclization protocol, we continued to investigate the application of the copper hydride catalytic system in the intermolecular C−H functionalization of benzimidazoles with allenes. We continued our studies by treatment of 1-methylbenzimidazole (**3a**) with cyclohexylallene (**4a**) using the optimized reaction conditions of CuH-catalyzed intramolecular C−H cyclization of benzimidazoles with allenes (Table 2). Fortunately, the desired product **5aa** was obtained in a moderate yield of 73% with 4:1 *E*/*Z* (entry 1 of Table 2). Subsequently, the effect of some other solvents was examined. Notably, toluene as a solvent instead of tetrahydrofuran (THF) led to a 93% yield with 15:1 *E*/*Z*

Table 2. Condition Optimization for CuH-Catalyzed Allylation of Benzimidazles[a]



entry	solvent	yield (%)[b]	*E*/*Z*[c]
1	THF	73	4:1
2	DCE	31	
3	CH_3CN	trace	
4	toluene	93	15:1
5	1,4-dioxane	0	
6	MTBE	0	

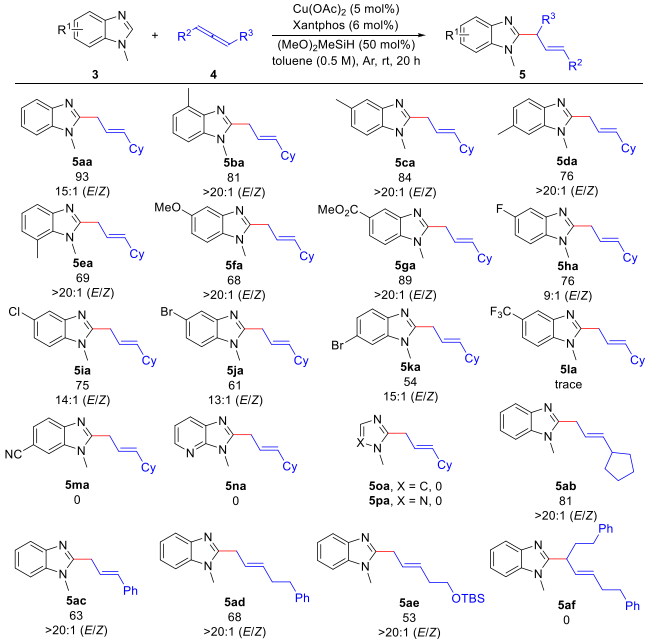
[a]Reaction conditions: compound **3a** (0.4 mmol), compound **4a** (0.6 mmol), $Cu(OAc)_2$ (5 mol %), Xantphos (6 mol %), $(MeO)_2MeSiH$ (50 mol %), and solvent (2 mL), at room temperature (rt) under argon for 20 h. [b]Isolated yield. [c]*E*/*Z* isomers was isolated and determined by ^1H nuclear magnetic resonance (NMR) spectroscopy.

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(entry 4 of Table 2). Then, the effect of other reagents was investigated; unfortunately, they did not exhibit better activity (see the Supporting Information for details).

We next explored the scope of this allylation protocol by examining various 1-methylbenzimidazoles (Scheme 3). The

Scheme 3. Substrate Scope of CuH-Catalyzed Allylation of Benzimidazoles[a,b,c]

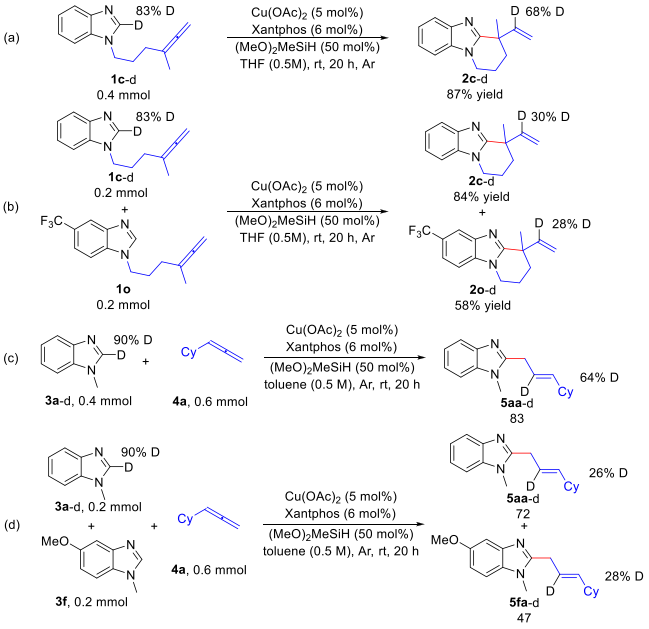


[a]Reactions were carried out with reaction conditions in entry 4 of Table 2. [b]Isolated yield. [c]E/Z isomers was determined by crude ^{1}H NMR spectroscopy.

reaction proceeded efficiently with 1-methylbenzimidazoles bearing electron-donating and electron-withdrawing groups, providing the corresponding products in moderate to good yields with good *E* isomer selectivity (**5aa**−**5ka**). 1-Methylbenzimidazoles bearing a methyl substituent at different positions afforded the corresponding products in 69−84% yields with good *E* isomer selectivity (**5ba**−**5ea**). 1-Methylbenzimidazoles containing a 5-methoxy group gave the product in a moderate yield of 68% (**5fa**). Interestingly, an ester-containing substrate could provide the desired product in an 89% yield (**5ga**). Substrates containing halogen groups (e.g., F, Cl, and Br) afforded the corresponding products in moderate yields (**5ha**−**5ka**). Unfortunately, Other substrates bearing electron-withdrawing groups (e.g., CF_3 and CN) were not suitable in this reaction (**5la** and **5ma**). N-containing heterocycles, imidazoles, or triazoles could not provide the corresponding products (**5na**−**5pa**). Subsequently, we proceeded to explore the scope of allenes under the optimized reaction conditions. Cyclohexylallene and cyclopentylallene could provide the corresponding products in good yields (**5aa** and **5ab**). Phenylallene provided the desired product in a moderate yield of 63% (**5ac**). Linear alkyl allenes were also tolerated in this reaction, providing the corresponding products in moderate yields (**5ad** and **5ae**). Unfortunately, internal allene was unsuitable for this allylation protocol (**5af**).

To obtain further insight into this CuH-catalyzed C−H allylation of benzimidazoles, some mechanistic experiments were performed (Scheme 4). Initially, C2-deuterated substrate

Scheme 4. Mechanistic Experiments

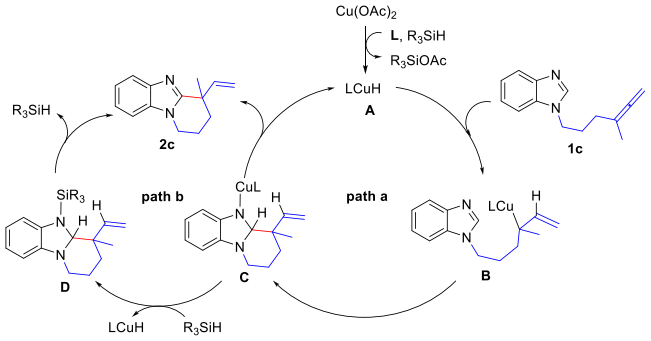


1c-*d* was reacted under the standard conditions of the cyclization protocol (equation a). To our delight, significant deuterium incorporation was observed at the alkene group of compound **2c**-*d*, which was different from the previous classic copper hydride catalytic mechanism.[10,11] Subsequently, a competition experiment between two different substrates (**1c**-*d* and **1o**) was performed under the standard conditions of the cyclization protocol (equation b). Interestingly, two deuterated products were observed. These results revealed that there was a deuterium-scrambling distribution. Then, C2-deuterated substrate **3a**-*d* was reacted under the standard conditions of the allylation protocol, and significant deuterium incorporation was observed at the alkene group of **5aa**-*d* (equation c). A competition experiment between two different substrates (**3a**-*d* and **3f**) under the standard conditions of the allylation protocol provided two deuterated products (equation d). These observations suggested that the hydrogen atom at the C2 position of benzimidazoles could participate in the formation of copper hydride species, which means that this reaction follows a different catalytic mechanism compared to the previous classic catalytic mechanism.[10,11]

On the basis of the above results, a possible catalytic mechanism for the cyclization protocol was proposed (Scheme 5). The first step would be the formation of LCuH species (**A**) with the assistance of $(MeO)_2MeSiH$ and phosphine ligand. The *in situ* generated LCuH species (**A**) undergoes a hydrocupration of allene **1c** to generate allyl copper intermediate **B**, which undergoes an intramolecular addition to the C=N bond to afford cyclization intermediate **C**. Then, the interaction of this copper nitride intermediate **C** with the hydrogen atom at the C2 position of benzimidazoles via a *β*-hydride elimination regenerates the LCuH species (**A**) to fulfill the catalytic cycle and provides the desired product **2c** (path a). However, in mechanistic studies, the reaction of C2-deuterated substrate **1c**-*d* provided the product **2c**-*d* in a 68% deuterated ratio, which was 15% lower than substrate **1c**-*d*. This result suggested that there perhaps exists a secondary catalytic cycle, which allows for generation of the intermediate **D** and LCuH species (**A**) through the interaction of copper

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Scheme 5. Proposed Mechanism for CuH-Catalyzed Cyclization of Benzimidazoles with Allenes



nitride intermediate **C** with $(MeO)_2MeSiH$ (path b). Next, a net loss of $(MeO)_2MeSiH$ provides the desired product **2c**. On the basis of our studies, we envisaged that the catalytic mechanism for the allylation protocol might follow a similar mechanism as found for the cyclization protocol (see the Supporting Information for details). The intermolecular addition of compound **3a** might undergo a process that proceeds through a six-membered transition state **A** (**TSA**) in which the cyclohexyl group occupies the exocyclic axial position, affording the *E* isomer intermediate **C**. The corresponding transition state **B** (**TSB**) affording the minor *Z* isomer intermediate **C** would require the cyclohexyl group to occupy the endocyclic axial position, resulting in a steric repulsion between the cyclohexyl group and benzimidazole, which might explain the *E* isomer selectivity.

In conclusion, we have developed the first example of copper-hydride-catalyzed intra- and intermolecular C−H functionalization of benzimidazoles with allenes, providing an atom economy protocol for the synthesis of various benzimidazole derivatives. In terms of green and sustainable chemistry, the copper salt catalyst and mild reaction conditions provide a particularly useful and energy-efficient tool for the C−H functionalization of benzimidazoles. The new copper hydride catalytic mechanism contributes more possibilities to modern organic synthesis.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.orglett.1c02346.

Experimental details and characterization of the products and starting materials (PDF)

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Notes

The authors declare no competing financial interest.

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